CRT Question Chapter 5

1. **What is the purpose of a loop structure?**

To automate repetitive tasks. This improves code efficiency, and cleans up codes that would otherwise use multiple lines to find specific conditions.

2. **Explain the difference between a *while* statement and a *do-while* statement.**

While loops are used when the loop might not need to run at all.
do-while loops are useful when you need to ensure the code block runs at least once, such as asking for user input and then validating if the input is correct. Only repeating the prompt if it is invalid.

3. **Input Validation loop**

NumbersSum review

4. **What is an infinite loop?**

An infinite loop is a loop that never stops running because its condition is always true or it has no proper end.

List two types of errors that can lead to an infinite loop.

- The loop control variable is not updated, forgetting to change the variable that controls the loop condition.
- Incorrect loop conditions happen when the condition is written wrong and that statement never becomes false.

  OVERFLOW - A condition that occurs when a number is too large to be stored in a specified number of bits.

5. **How many times will the do-while loop execute?**

The loop executes 60 times. Because after its 60th iteration, x becomes 120 and the condition x < 120 becomes false. Stopping the loop.

6. **What initial value of x would make the loop infinite?**

Any value where x < 120

7. **Compare and contrast counters and accumulators. List two uses for each.**

Counters - A variable that is incremented by a fixed value.

- Counts occurances
- Usually +1 each time

Uses

- Counting how many inputs meet a specific condition (ex. How many numbers are positive)
- Teaching number of loop iterations (ex. Repeat something 10 times)

Accumulator - A variable that is incremented by varying amounts.

- Adds values together
- + variable amount

Uses

- Calculate a total (ex. Sum of all numbers entered)
- Finding an average (ex. By accumulating total, then dividing by count)

8. **Write a *for* statement that sums the integers from 3 to 10, inclusive.**

```
int sum = 0;

for (int i = 3; i <= 10; i++) {
        Sum += i;
}
```

9. **List two factors that should be considered when determining which loop structure to choose.**

- Whether the loop must run at least once.
- Type of control needed for the loop. (known number of iterations, or until a condition is met)

11. Consider the following assignment:

a) 10
b) “my “
c) “my string.”
d) “MY STRING.”
e) “my string.”